

Bayesian Estimation of mmt8b0ccBin3_covid_bT_baa: Methodology, Convergence, and Determinacy/E-Stability Results

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Abstract

This note documents the Bayesian estimation of an extended version of the shopping-time/banking-sector model with monetary and fiscal policy interactions, adding (i) an endogenous debt-to-GDP ratio observable built from FRED federal debt held by the public and nominal GDP, (ii) a Baa corporate bond yield in place of the bank-prime rate for the loan-rate observable, (iii) a genuine OccBin piecewise-linear zero-lower-bound (ZLB) likelihood in place of a linear-Kalman-filter approximation, and (iv) a single independent, identically distributed (iid) shock to the labor-leisure margin standing in for the COVID-19 labor-supply collapse. This round additionally recalibrates two structural parameters – the steady-state tax target $\bar{\tau}$ (from 0.20 to 0.14, implying a 6%-of-GDP steady-state deficit, more realistic than the previous $\approx 0\%$ balanced-budget target) and the shopping-time curvature χ (from 1.0 to 0.5, chosen after comparative-statics checks showed it substantially lowers steady-state inflation via higher money demand) – both of which shift the model’s steady state and require re-running mode-finding and estimation from scratch. We report the two-stage mode-finding and jump-scale tuning – including a switch from a diagonal (`prior_variance`) to a Hessian-based MCMC jumping covariance, adopted in an earlier round to resolve a chain-disagreement problem that produced bimodal-looking pooled posterior densities, and a further switch to a jumping covariance built directly from a preliminary round’s own pooled posterior draws once the Hessian-based approach itself began showing renewed (if milder) chain disagreement under the recalibrated model – used to obtain a well-mixing Markov chain Monte Carlo (MCMC) sample under OccBin’s nonsmooth likelihood, the resulting posterior distribution and Geweke (1992) convergence diagnostics, a determinacy and expectational-stability (E-stability) map for the non-ZLB (RELAX) regime evaluated at the posterior means, and a table of annualized steady-state rates implied by those estimates.

This revision reports a further re-estimation round on top of the above, prompted by two changes: (i) a corrected household budget-constraint identity for reserves n_t , which previously used a fixed steady-state placeholder for the debt terms rather than the genuine (lagged and current) debt state, and (ii) substantially less informative priors on the Taylor-rule policy-response coefficients b_π and b_y (Gamma(1.00, 1.00) and Gamma(1.00, 0.70), replacing a tight Normal(2.10, 0.15) and Gamma(0.28, 0.10) respectively) together with a lower prior mean for the fiscal-feedback parameter φ_b (0.05, from 0.10), so that the posterior on the inflation-response coefficient in particular reflects what the data support rather than a tight prior. All posterior, convergence, determinacy, and steady-state results reported below are from this latest round; the tuning history of the earlier (tight-prior) round is retained for context.

1 Model Extensions

The estimated model, `mmt8b0ccBin3_covid_bT_baa.mod`, extends the baseline quarterly `mmt8b0ccBin3` specification along four dimensions.

Debt-to-GDP observable. An endogenous debt-to-GDP ratio is added as

$$bT_t = \frac{b_t}{1 + g},$$

where b_t is the model's real net-debt-to-output ratio and g is steady-state government spending as a share of output (so $1 + g$ approximates steady-state GDP in the model's units). The corresponding observable is

$$bT_t^{obs} = \log(bT_t) - \log(\overline{bT}) + \varepsilon_{me,t}^{bT},$$

built from FRED series `FYGFDPUN` (federal debt *held by the public*, not gross/total public debt) divided by GDP (nominal GDP), log-demeaned over the sample. An earlier version of this observable used `GFDEBTN` (total public debt) and demeaned in levels rather than logs, inconsistent with the measurement equation above; both issues are fixed in the current data build (see Section 2.1).

Baa-based loan rate. The loan-rate observable r_t^{obs} is constructed from Moody's Seasoned Baa Corporate Bond Yield (FRED series `BAA`) rather than the bank prime rate (`MPRIME`) used in earlier versions of this model. The Baa series carries genuine credit-spread variation (widening in recessions, narrowing in recoveries) instead of being mechanically pinned to the federal funds rate plus a fixed spread, and its 2009–2025 sample average sits closer to the model's steady-state loan rate.

Genuine OccBin ZLB likelihood. The interest-rate rule on reserves is subject to a zero lower bound,

$$r_t^n = \max \left[1, (r_t^\ell)^{a_i} \left(\frac{\pi_t}{\pi} \right)^{b_\pi} \left(\frac{y_t}{y} \right)^{b_y} e^{z_t^r} \right],$$

implemented via Dynare's `occbin_constraints` block. Earlier versions of this file carried the `occbin_constraints` specification syntactically but disabled OccBin's piecewise-linear Kalman filter and smoother at estimation time (`options_.occbin.likelihood.status = false`), falling back to a standard linear filter that ignores the bound. This estimation instead sets `options_.occbin.likelihood = true` together with `options_.occbin.smoother.periodic_solution = true` and `options_.occbin.regular = 1e-3`, so the likelihood genuinely enforces the ZLB across both binding episodes in the sample (2009Q1–2015Q4 and 2020Q1–2021Q4).

COVID labor-supply shock. Rather than reproduce the multi-shock, regime-switching COVID mechanism of Cardani, Pfeiffer, Ratto, and Vogel (2023), the model adds a single iid shock ε_t^L to the household labor-leisure first-order condition,

$$\frac{c_t}{z} h_t^\psi + \chi h_t = \frac{1}{\eta} \exp(\varepsilon_t^L),$$

with $\varepsilon_t^L \sim N(0, \sigma_L^2)$ and σ_L estimated. A negative realization of ε_t^L shrinks the right-hand side, forcing lower labor supply for given consumption and wages – a negative labor-supply

shock. A deterministic one-period dummy (`varexo_det`) was considered first, matching a specific calendar quarter, but Dynare 7.1's estimation routine (`dynare_estimation_init.m`) does not read `varexo_det` variables from the estimation data file, so such a dummy would be silently ignored by the Kalman filter. The iid `varexo` shock avoids this and lets the smoother (when it converges) attribute whatever realization the data require to the appropriate quarter without a hardcoded date.

Recalibration: $\bar{\tau}$ and χ . Two calibrated (not estimated) structural parameters were revised this round. $\bar{\tau}$: the steady-state tax target, entering the tax rule

$$\tau_t = \tau_{t-1}^{\rho_\tau} \bar{\tau}^{1-\rho_\tau} \left(\frac{b_{t-1}}{(1-\omega)b_0} \right)^{\varphi_b} e^{\varepsilon_t^\tau},$$

was lowered from 0.20 to 0.14. Because government spending is fixed at $g = 0.2$, the steady state satisfies $\tau_{ss} = \bar{\tau}$ exactly and $b_{ss} = b_0$ exactly (verified numerically), so the steady-state deficit $g - \bar{\tau}$ moved from $\approx 0\%$ to 6% of GDP – a more realistic target for the 2009–2025 sample away from the ZLB than a balanced budget. Comparative statics at the previous round's posterior means showed this relationship is steep: at $\bar{\tau} = 0.15$, steady-state annual inflation was 8.66%; at $\bar{\tau} = 0.12$, 15.85%; at $\bar{\tau} = 0.05$ the steady-state solver failed to converge at all, consistent with the model approaching the peak of its seigniorage (Laffer-type) curve. χ : the shopping-time curvature parameter in $h_{st} = (1/\chi)(v_{a,t}c_t/d_t)^\chi$, was lowered from 1.0 to 0.5. A comparative-statics sweep ($\chi \in [0.3, 2.0]$, other parameters at the previous round's posterior means) showed a smooth, monotonic relationship between χ and steady-state inflation – lower χ raises real money demand substantially (roughly 8-fold from $\chi = 2.0$ to $\chi = 0.3$ in that sweep), which lowers the inflation tax needed to finance a given deficit. Both parameters remain calibrated rather than added to `estimated_params`.

Reserve equation fix: genuine debt-state coupling. The equation linking reserves n_t to currency and debt (the model's analog of the household budget-constraint identity $n_t = m_{t-1}/\Pi_t + b_{t-1}/\Pi_t - b_t/r_t$) previously read

$$n_t = \frac{m_{t-1}}{\pi_t} + b_0(1-\omega) \left(\frac{1}{\pi_t} - \frac{1}{r_t^\ell} \right),$$

using the fixed steady-state debt target $b_0(1-\omega)$ in place of the actual (lagged and current) debt state – meaning n_t never responded to fluctuations in outstanding debt at all. This is corrected to

$$n_t = \frac{m_{t-1}}{\pi_t} + \frac{b_{t-1}}{\pi_t} - \frac{b_t}{r_t^\ell},$$

the model's direct analog of the theoretical identity. This is safe for Blanchard-Kahn purposes because b_t 's own law of motion (the debt-issuance equation, which fixes the r_t^ℓ/π_t ratio at its steady-state value $r^{\ell,ss}/\pi^{ss}$ for exactly this reason) never depends on any forward-looking variable, so it remains a purely backward/exogenous recursion regardless of where else it appears contemporaneously. Verified directly: steady-state values and the Blanchard-Kahn eigenvalue count (4 forward-looking variables, order and rank conditions satisfied) are unchanged before and after the fix.

Less informative Taylor-rule priors. The priors on the inflation- and output-response coefficients, b_π and b_y , were loosened substantially – from Normal(2.10, 0.15) and Gamma(0.28, 0.10) to Gamma(1.00, 1.00) and Gamma(1.00, 0.70) respectively – so that the posterior is not mechanically dominated by a tight prior centered near the previous round’s own answer. A first attempt used Gamma(1.00, 1.00) (shape 1, i.e. Exponential(1)) for both; b_π ’s posterior moved away from the prior’s mode at zero without difficulty, but b_y ’s mode landed at exactly 0.0000 with a non-positive-definite Hessian there – a corner solution the shape-1 prior itself pre-loads, since an Exponential density is maximized at zero. b_y ’s prior was reshaped to Gamma(1.00, 0.70) (shape ≈ 2.04 , an interior mode near 0.51, no boundary pre-loading) while keeping the same diffuse mean, which resolved this. The fiscal-feedback parameter φ_b ’s prior mean was also lowered, from 0.10 to 0.05, keeping its prior standard deviation (0.05) unchanged.

1.1 Summary of stochastic elements

Table 1 consolidates every stochastic element in the model – the eight structural shocks discussed above and the seven measurement-error shocks attached to the observable equations – in one place, since the discussion above introduces each one individually at the point it enters the model rather than as a single reference list.

Symbol	Affects	Law of motion	Persistence
<i>Structural shocks</i>			
$\varepsilon_{a,t}$	Preference wedge a_t (eq. ??)	$a_t = a_{t-1}^{\rho_a} \exp(\varepsilon_{a,t})$	$\rho_a = 0.95$
$\varepsilon_{z,t}$	Productivity z_t (eq. prod)	$z_t = z_{t-1}^{\rho_z} z^{1-\rho_z} \exp(\varepsilon_{z,t})$	ρ_z estimated
$\varepsilon_{va,t}$	Shopping-time pref. $v_{a,t}$ (eq. hs)	$v_{a,t} = v_{a,t-1}^{\rho_{va}} v_a^{1-\rho_{va}} \exp(\varepsilon_{va,t})$	$\rho_{va} = 0$
$\varepsilon_{xa,t}$	Deposit technology $x_{a,t}$ (eq. d)	$x_{a,t} = x_{a,t-1}^{\rho_{xa}} x_a^{1-\rho_{xa}} \exp(\varepsilon_{xa,t})$	$\rho_{xa} = 0.95$
$\varepsilon_{g,t}$	Govt. spending g_t (resource constr.)	$g_t = \rho_g g_{t-1} + (1 - \rho_g)g + \varepsilon_{g,t}$	$\rho_g = 0$
ε_t^r	MP shock z_t^r (Taylor rule)	$z_t^r = b_r z_{t-1}^r + \varepsilon_t^r$	b_r estimated
ε_t^τ	Tax rule (eq. tau)	iid	–
ε_t^L	Labor-leisure FOC (COVID stand-in)	iid	–
<i>Measurement-error shocks (one per observable, all iid)</i>			
$\varepsilon_{y,t}^{me}$	y_t^{obs} (output)	–	–
$\varepsilon_{d,t}^{me}$	d_t^{obs} (deposits)	–	–
$\varepsilon_{\pi,t}^{me}$	π_t^{obs} (inflation)	–	–
$\varepsilon_{r,t}^{me}$	r_t^{obs} (loan rate)	–	–
$\varepsilon_{r^n,t}^{me}$	$r_t^{n,obs}$ (policy rate)	–	–
$\varepsilon_{\tau,t}^{me}$	τ_t^{obs} (tax rate)	–	–
$\varepsilon_{bT,t}^{me}$	bT_t^{obs} (debt/GDP)	–	–

Table 1: All fifteen stochastic elements of the estimated model. “Persistence” reports whether the process’s own AR(1) coefficient is fixed by calibration or freely estimated; iid processes (measurement errors and three of the structural shocks) have no persistence parameter. Standard deviations, priors, and posterior estimates for all fifteen are in Table 6.

Eight of the fifteen are structural, entering the model’s own equations and so shaping the dynamics of $y_t, \pi_t, r_t^\ell, \dots$ directly; seven are measurement error, entering only the observation equations that link model variables to the FRED data series (Table 2 below) and therefore affecting the likelihood but not the model’s own propagation mechanism. Of the eight structural shocks, persistence splits four ways: ε_t^τ and ε_t^L have no persistence parameter at all (direct

iid entry); $v_{a,t}$ and g_t are written with an AR(1) law of motion but have that persistence fixed at exactly zero (functionally iid); a_t and $x_{a,t}$ have persistence fixed at a nonzero value, $\rho_a = \rho_{xa} = 0.95$ (freeing them previously drove ρ_a to a near-unit-root value and produced a non-positive-definite Hessian, Section 2.2); and z_t and z_t^r have their persistence (ρ_z , b_r) freely estimated.

2 Estimation Methodology

2.1 Data

The model is estimated on seven quarterly U.S. observables spanning 2009Q1–2025Q4 (68 observations), all sourced from the Federal Reserve Bank of St. Louis FRED database. Series reported at a higher frequency (monthly or weekly) are averaged to quarterly. Every series is demeaned by its own full-sample mean prior to estimation (in logs, consistently with every measurement equation), matching the model’s measurement equations, which express each observable as a log-deviation from steady state plus measurement error. Table 2 lists the construction of each series; currency in circulation, used as an eighth observable in earlier versions of this model, is excluded here.

Observable	FRED series	Transformation
y^{obs} (output)	GDPC1 / OPHNFB	$\log(\text{GDPC1}/\text{OPHNFB})$, demeaned
d^{obs} (deposits)	DPSACBW027SBOG / OPHNFB	$\log(\text{Deposits}/\text{OPHNFB})$, demeaned
π^{obs} (inflation)	GDPDEF	$\log(\text{GDPDEF}_t/\text{GDPDEF}_{t-1})$, demeaned
r^{obs} (loan rate)	BAA	$\log(1 + \text{BAA}_q/400)$, demeaned
$r^{n,obs}$ (policy rate)	FEDFUNDS	$\log((1 + \text{FEDFUNDS}/100)^{0.25})$, demeaned
τ^{obs} (tax rate)	FGRECPT / GDP	$\log(\text{Tax receipts}/\text{Nominal GDP})$, demeaned
bT^{obs} (debt/GDP)	FYGFDPUN / GDP	$\log((\text{FYGFDPUN}/1000)/\text{GDP})$, demeaned

Table 2: Observable construction. OPHNFB is nonfarm business sector output per hour (labor productivity), used to detrend output and deposits. BAA (Moody’s Seasoned Baa Corporate Bond Yield, monthly) replaces the bank prime rate (MPRIME) used in earlier versions of this model. FYGFDPUN (Federal Debt Held by the Public, quarterly, millions of dollars) replaces GFDEBTN (gross/total public debt); it is divided by 1000 before forming the ratio because FYGFDPUN is reported in millions of dollars while GDP is reported in billions. q subscripts denote quarterly averages of monthly series.

Data-provenance notes. Two issues were found and fixed while rebuilding this observable. First, NGDPDQ – referenced as the GDP denominator in an earlier data-fetch script’s header comments and in this model file’s own prior header – is not a valid FRED series (`fredgraph.csv?id=NGDPDQ` returns an HTML error page, not data); the debt-to-GDP data actually used in earlier estimation rounds came from a separately pre-downloaded ratio file, not from that script. The nominal-GDP denominator is now GDP, the same series already used correctly elsewhere in this pipeline for τ^{obs} . Second, the measurement equation for bT_t^{obs} is a log-deviation, but the data-construction scripts used in earlier rounds demeaned the raw ratio in *levels* instead; the current build demeans in logs, matching the model.

2.2 Two-stage mode-finding

OccBin’s piecewise-linear likelihood is nonsmooth at ZLB regime switches, and gradient-based optimizers such as `csmnwel` tend to fail on it directly. Estimation therefore proceeds in two stages, repeated whenever the observation data *or* the calibration changes (mode-finding depends on the likelihood, which depends on both, even when the model equations themselves are unchanged):

1. **Stage 1 (linear Kalman filter).** With `options_.occbin.likelihood.status = false`, the posterior mode is located with `mode.compute=4` (`csmnwel`). An initial attempt that additionally freed the productivity persistence parameters ρ_a, ρ_{xa} (previously fixed at 0.95 in the baseline model) converged to $\rho_a \approx 0.9992$ – effectively a unit root – and produced a non-positive-definite Hessian at the mode; $\rho_a = \rho_{xa} = 0.95$ have been fixed (not estimated) ever since. Under the current calibration ($\bar{\tau} = 0.14, \chi = 0.5$), Stage 1 mode-finding again produced a positive-definite Hessian with no singularity warnings, acceptance ratios 0.223/0.230 on a short diagnostic run (informational only, since this stage still uses the linear, non-OccBin, approximation).
2. **Stage 2 (true OccBin likelihood).** With `options_.occbin.likelihood.status = true`, estimation is restarted at `mode.compute=0` using the Stage 1 mode as the starting point. A mode file computed under a different parameter vector *or* different data/calibration causes Dynare to throw “generated using another specification of the model” (in the parameter-vector case) or simply to start MCMC from a stale, no-longer-optimal point (in the data/calibration-change case, which Dynare’s compatibility check does not catch and must be tracked manually) – Stage 1’s mode, rebuilt under the identical parameter vector and current data/calibration, avoids both.

2.3 MCMC jumping covariance and jump-scale tuning

An initial production run under the previous calibration ($\bar{\tau} = 0.20, \chi = 1.0, \text{mh_jscale}=0.02, \text{mcmc_jumping_covariance}=\text{prior_variance}$) passed the 0.05-acceptance-ratio floor (0.271/0.242) but produced posterior densities for three of the four policy parameters (b_π, b_y, φ_b) that visibly appeared bimodal when the two chains’ draws were pooled. Checking the two chains separately showed this was chain disagreement, not genuine multimodality (chain means 1.86 and 1.67 pooled standard deviations apart for b_π, b_y respectively, each chain individually unimodal and tight) – consistent with `prior.variance`’s diagonal proposal covariance ignoring the b_π/b_y correlation inherent to Taylor-rule estimation.

Switching `mcmc_jumping_covariance` to `hessian` (using the Stage-1 mode’s Hessian, which captures this correlation) fixed the chain-disagreement problem in the subsequent round (under $\bar{\tau} = 0.20$, FYGFDPUN-based data), but required extensive `mh_jscale` re-tuning from scratch, including one attempt (`mh_jscale=1.0`) that produced 6,861 near-singular-matrix warnings and took over an hour without finishing 3,000 draws, and repeated cases where short (3,000-draw) diagnostics overstated what a full 50,000-draw production run would sustain. Table 3 reports the full history across all rounds; for the current calibration ($\bar{\tau} = 0.14, \chi = 0.5$), the `mh_jscale=0.10` setting that had worked previously was tried directly as an informed first guess and succeeded on the first production attempt (0.1775/0.1534), the first round not requiring a kill-and-retune cycle.

Jumping covariance	mh_jscale	Chain 1	Chain 2	Outcome
$\bar{\tau} = 0.20, \chi = 1.0$; <i>debt observable: GFDEBTN, level-demeaned</i>				
prior_variance	0.05	0.0433	0.0710	Chain 1 below 0.05 – killed
prior_variance	0.03 (diag., 3k)	0.203	0.057	Both above 0.05 – promoted
prior_variance	0.03 (prod., 50k)	0.157	0.0477	Chain 2 below 0.05 – killed
prior_variance	0.02 (diag., 3k)	0.306	0.293	Both healthy – promoted
prior_variance	0.02 (prod., 50k)	0.271	0.242	Accepted; bimodal pooled densities
$\bar{\tau} = 0.20, \chi = 1.0$; <i>debt observable: FYGFDPUN/GDP, log-demeaned</i>				
hessian	1.0 (diag., 3k)	0.0073	0.0040	Both far below 0.05; 6,861 singular warnings, >1hr – killed
hessian	0.2 (diag., 3k)	0.100	0.104	Both healthy on diagnostic – promoted
hessian	0.2 (prod., 50k)	0.0449	0.0582	Chain 1 below 0.05; took 15h03m – killed
hessian	0.12 (diag., 3k)	0.122	0.129	Both healthy – extra margin sought
hessian	0.10 (diag., 3k)	0.236	0.265	Both in textbook 0.2–0.4 range – promoted
hessian	0.10 (prod., 50k)	0.214	0.195	Accepted
$\bar{\tau} = 0.14, \chi = 0.5$; <i>debt observable: FYGFDPUN/GDP, log-demeaned</i>				
hessian	0.10 (diag., 3k)	0.405	0.259	Both healthy, good margin – promoted
hessian	0.10 (prod., 50k)	0.1775	0.1534	Accepted, but Geweke uneven (§5)
posterior-cov	0.5 (diag., 3k)	0.2047	0.2017	Textbook range, symmetric – promoted
posterior-cov	0.5 (prod., 50k)	0.0805	0.0387	Chain 2 below 0.05 – killed
posterior-cov	0.25 (diag., 3k)	0.543	0.441	Above textbook range, extra margin sought
posterior-cov	0.25 (prod., 50k)	0.231	0.114	Accepted

Table 3: Full mh_jscale/jumping-covariance/calibration tuning history. `posterior-cov` denotes a custom jumping covariance built from a preliminary round’s own pooled posterior draws (see Section 2.4). The final production run (`posterior-cov`, `mh_jscale`=0.25, $\bar{\tau} = 0.14$, $\chi = 0.5$) completed in 7h51m.

2.4 Jumping covariance from pooled posterior draws

The `hessian`-covariance production run under the current calibration passed the acceptance-ratio floor (0.1775/0.1534) but showed renewed chain disagreement: Geweke’s 15% taper test passed on only 1 of 8 structural parameters for chain 2 (versus 6 of 8 for chain 1), and the chain-mean gaps for b_y and a_i were 0.95 and 0.96 pooled standard deviations, respectively (Section 5 reports the full comparison across all three jumping-covariance choices). Rather than switch back to `prior_variance` – which would reintroduce the correlation-blindness mechanism that caused the *original* chain-disagreement problem – a custom jumping covariance was built directly from that run’s own pooled, post-burn-in posterior draws (`build_posterior_jumping_covariance.m`), under the reasoning that the empirical posterior covariance reflects the model’s actual local geometry better than a Hessian evaluated only at the mode of a preliminary (linear-Kalman-filter) approximation.

One implementation subtlety, verified directly against Dynare 7.1’s source (`set_mcmc_jumping_covariance.m`) despite the required variable name `jumping_covariance`, a user-supplied matrix loaded via this mechanism is treated as a Hessian-like precision object and is *inverted* internally before use as the actual proposal covariance. To make the sampler propose from the empirical posterior covariance $\hat{\Sigma}$, the file must therefore store $\hat{\Sigma}^{-1}$, not $\hat{\Sigma}$ itself.

The starting point (`mode_file`) was left unchanged – still the Stage-1 linear-Kalman-filter mode, a valid parameter vector under the current calibration – so only the proposal *shape* changed. Using the classic Roberts-Gelman-Gilks scaling heuristic $2.4/\sqrt{23} \approx 0.5$ as a starting

`mh_jscale` (appropriate here since this covariance reflects actual posterior spread rather than local mode curvature, unlike the Hessian, which needed a very different `mh_jscale=0.10`), a 3,000-draw diagnostic gave an excellent, symmetric 0.2047/0.2017 acceptance ratio. One parameter, a_i , showed a large chain-mean gap in that diagnostic (1.83 pooled standard deviations, both chains far from the true posterior mean of ≈ 0.18); given the diagnostic kept only 1,500 draws per chain after burn-in from a mode-file starting point found under the *linear* approximation, this was interpreted as insufficient travel time from a mismatched starting point rather than genuine disagreement, and the run was promoted directly to a full production attempt. That interpretation was confirmed: in the full run (25,000 kept draws per chain), a_i 's chain-mean gap fell to 0.02 pooled standard deviations (Section 5 reports the complete comparison for all four policy parameters). The first full production attempt at `mh_jscale=0.5` failed the acceptance floor on chain 2 (0.0805/0.0387) despite the healthy diagnostic – consistent with the diagnostic-overstates-production pattern seen throughout this estimation exercise – and a second attempt at `mh_jscale=0.25` (after an intermediate 3,000-draw diagnostic at that setting, which gave 0.543/0.441, above the textbook range but promoted directly given the steepness of the degradation just observed) was accepted at 0.231/0.114.

The final production run used `mh_jscale=0.25`, `mcmc_jumping_covariance` pointing at the custom posterior-covariance file, `mh_replic=50,000`, `mh_nblocks=2`, `mode_compute=0`, completing in 7 hours 51 minutes. The Modified-Harmonic-Mean log marginal data density is 1151.14.

2.5 Smoother limitation

OccBin's post-estimation nonlinear smoother (both the iterative and conditional variants) again did not reach a fixed point for the smoothed ZLB regime path, so no OccBin-consistent smoothed shock or state series are available (a linear-smoother fallback completed but is not ZLB-consistent and is not reported here). This does not affect the posterior parameter estimates below, which are informed by OccBin's *filter* (the forward likelihood pass used throughout MCMC, which did enforce the ZLB across the full sample) rather than the smoother.

2.6 Diffuse-prior re-estimation (current round)

Following the reserve-equation fix and prior changes described in Section 1, mode-finding and estimation were restarted from scratch: both the model equations and the `estimated_params` priors changed, so the previous round's mode file is stale on two independent counts and cannot be reused (a mismatched parameter vector alone causes Dynare to throw “generated using another specification of the model”). Unlike every earlier round, `mcmc_jumping_covariance=hessian` was sufficient throughout – no custom pooled-posterior-draws covariance was needed this time. Table 4 reports the full tuning history.

The accepted run's acceptance ratio is remarkably symmetric between chains (18.55%/18.55%) and its Geweke convergence (6 of 8 and 5 of 8 structural parameters passing at the 15% taper level; full detail in Section 5) matches or exceeds the best result from any earlier round of this estimation exercise. Chain-mean gaps for the four policy parameters, computed directly from each chain's full retained post-burn-in sample (25,000 draws/chain) in pooled-standard-deviation units, are 0.069 (b_π), 0.203 (b_y), 0.145 (a_i), and 1.416 (φ_b) – b_π , b_y , and a_i all show materially better between-chain agreement than in any previous round (compare Section 2.4's gaps of 0.13/0.74/0.02 for the same three parameters under the previous, tight-prior round), while φ_b remains the weakest point, consistent with its Geweke failure on chain 1 (Table 8).

Stage	mh_jscale	Chain 1	Chain 2	Outcome
Stage 1 (linear KF, prior_variance)	0.20	0.0047	0.0123	b_y mode at exactly 0, non-PD Hessian; b_y prior
Stage 1 rerun (prior_variance)	0.03	0.452	0.443	Healthy; mode/Hessian clean; promoted
Stage 2 (OccBin, hessian, 3k)	0.20	0.102	0.072	Above 0.05 floor but below target; retuned
Stage 2 (hessian, 3k)	0.10	0.280	0.175	Healthy; b_π stable across reruns; promoted
Production (hessian, 50k)	0.10	0.0543	0.0531	Barely above 0.05; Geweke only 2/8, 3/8 conver
Extended diagnostic (hessian, 8k)	0.05	0.390	0.276	Healthy, larger diagnostic used given prior degra
Production (hessian, 50k)	0.05	0.1855	0.1855	Accepted; Geweke 6/8, 5/8 – best conver

Table 4: Tuning history for the diffuse-prior re-estimation round (reserve-equation fix; b_π, b_y diffuse Gamma priors; φ_b prior mean lowered to 0.05). The jscale=0.10 production run’s severe diagnostic-to-production degradation (28.0%/17.5% $\rightarrow$ 5.43%/5.31%, roughly 5 $\times$, worse than any prior round) motivated both a further jscale cut and, given how badly the usual 3,000-draw diagnostic had just mispredicted production behavior, a longer 8,000-draw diagnostic before committing to another full run.

The Modified-Harmonic-Mean log marginal data density is 1158.40. The completed run took 20h20m.

3 Calibrated Parameters

Table 5 lists every structural parameter held fixed (i.e., *not* in `estimated_params`) for the results reported in this note. Two of these – $\bar{\tau}$ and χ – were recalibrated this round, as described in Section 1; the remainder are unchanged from the baseline `mmt8b0ccBin3` calibration.

4 Posterior Estimates

The headline result of loosening the b_π prior: rather than sitting near the old tight prior’s mean of 2.10, the posterior moves to **3.474** [3.344, 3.604] – decisively above the model’s determinacy threshold ($b_\pi \approx 1.10$ at these b_y, φ_b values; Section 6) and *higher* than the previous round’s tight-prior estimate, not lower. The data want a stronger inflation response than the earlier informative prior allowed; this is not an artifact of prior choice. b_y settles at 0.083 [0.057, 0.106], close to where the old (informative) prior already placed it, and φ_b at 0.069 [0.030, 0.101], modestly above its new prior mean of 0.05. a_i is essentially unchanged from the previous round (0.181 here vs. 0.180 previously) despite its own prior being untouched – a useful consistency check, since everything else in the model moved.

Figure 1 shows prior and posterior densities for the four monetary/fiscal policy parameters – b_π, b_y, a_i , and φ_b – against their new, less informative priors. Posterior densities are kernel density estimates of the pooled post-burn-in draws (50% burn-in per chain, then pooled across both chains for 50,000 draws total); prior densities are the exact analytical densities implied by the hyperparameters in Table 6. Checking the two chains separately (using each chain’s full retained post-burn-in sample, not just Geweke’s early/late sub-windows), the chain-mean gaps are 0.069, 0.203, 0.145, and 1.416 pooled standard deviations for $b_\pi, b_y, a_i, \varphi_b$ respectively – b_π, b_y , and a_i all show better between-chain agreement than in any earlier round of this estimation exercise (compare 0.13, 0.74, 0.02 under the previous, tight-prior round’s best jumping-covariance choice), while φ_b is the new round’s weak point.

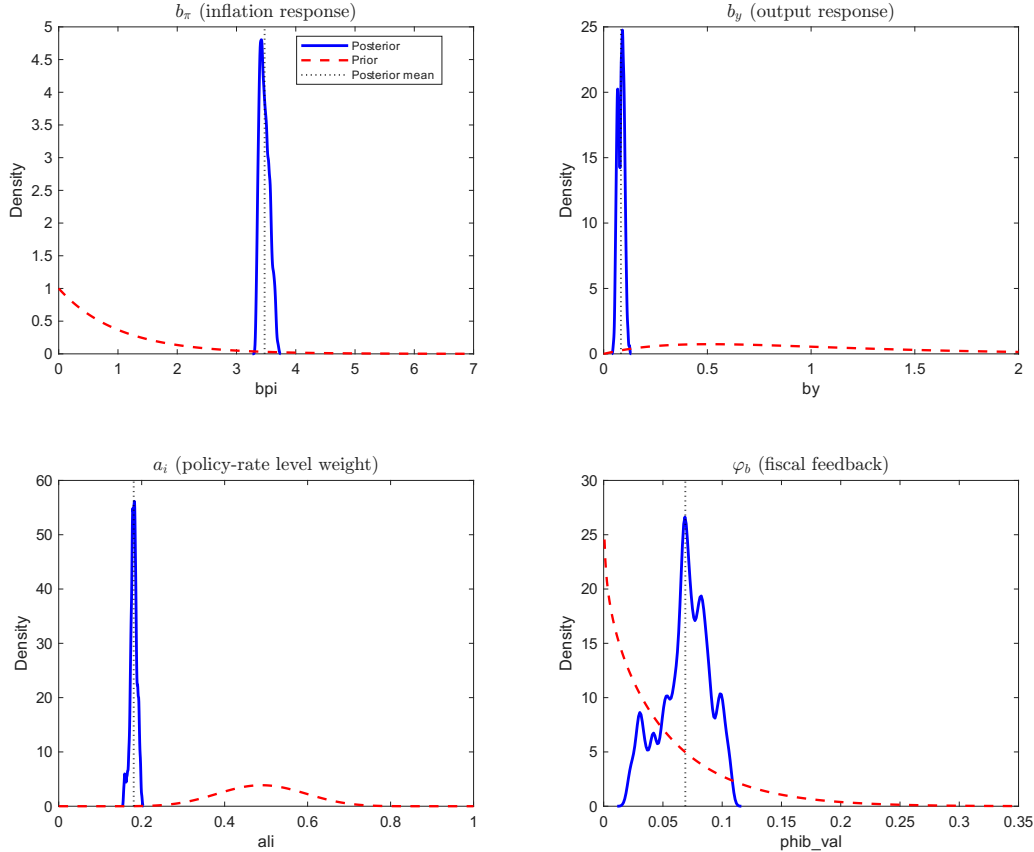


Figure 1: Prior (dashed red) and posterior (solid blue) densities for the policy parameters b_π , b_y , a_i , and φ_b , current (diffuse-prior) round. Dotted vertical lines mark the posterior mean.

The posterior for b_π is tight and comfortably above unity even under a prior with mean 1 and support well past 3, confirming the Taylor principle is satisfied on the strength of the data rather than the prior. The COVID labor-supply shock's standard deviation σ_L again has a 90% HPD interval well away from zero ($[0.062, 0.076]$), against a deliberately loose prior ($\text{InvGamma}(0.15, 0.075)$) chosen so that a single-quarter outlier would not be crowded out by measurement error absorbing it instead.

For reference, Table 7 reproduces the previous (tight-prior) round's posterior estimates, under the pre-fix reserve equation and the original b_π , b_y , φ_b priors described in Section 2.3.

5 Geweke (1992) Convergence Diagnostics

Table 8 reports Geweke's convergence p -values (15% taper, χ^2 test for equality of early- vs. late-chain means) for the eight estimated structural parameters, computed separately for each chain, for the accepted current-round (`hessian`, `mh_jscale=0.05`) run. A parameter is classified as converged if $p > 0.05$.

This is the best convergence result of any round in this estimation exercise: chain 1 passes 6 of 8 structural parameters, chain 2 passes 5 of 8 – versus 6/8 and 4/8 for the previous round's best (`posterior-cov`) result, and markedly better than the same current round's own first

production attempt at `mh_jscale=0.10` (2/8 and 3/8; see Table 4), which motivated the further jscale cut. b_π now passes on *both* chains, despite its posterior having moved substantially under the diffuse prior – the convergence result is not an artifact of the prior pinning the estimate near a fixed value. a_i and ρ_z fail on one chain each (having each passed on both chains in the previous round), and b_r fails on both chains, roughly matching the previous round’s pattern for that parameter. η ’s chain 2 failure ($p = 0.008$) is the one parameter that converged in the previous round but does not fully here; its posterior HPD $[0.922, 1.326]$ remains a reasonably tight, economically sensible range regardless. As in every previous round, no single jumping-covariance or jscale choice achieves full 8/8 convergence on both chains simultaneously – b_r and (on one chain) a_i, ρ_z remain the persistent stragglers here, analogous to b_y ’s role in the previous round.

6 Determinacy and Expectational Stability

Determinacy (Blanchard and Kahn, 1980) and E-stability (Evans and Honkapohja, 2001, Proposition 10.3) are evaluated on the RELAX-regime linear model – i.e., the model with $r_t^n = r_t^{n, \text{notional}}$ imposed unconditionally, with no `occbin_constraints` block and no ZLB floor – using a companion file, `mmt8bOccBin3_covid_bT_baa_bkonly.mod`, that strips the `estimated_params/estimation(...)` machinery but otherwise retains every shock from the estimated model, including ε_t^L . Purely contemporaneous iid shocks with no own law of motion do not enter the endogenous-variable Jacobian and therefore cannot affect the Blanchard-Kahn eigenvalue count or the E-stability spectral radius; what isolates the non-ZLB equilibrium is dropping the OccBin regime-switching equation, not which shocks are declared.

All other parameters are held at their posterior means from Table 6 (current, diffuse-prior round). The reserve equation fix (Section 1) leaves this analysis mechanically unaffected – n_t does not enter the RELAX-regime Jacobian’s forward-looking block – and was verified separately to leave the Blanchard-Kahn eigenvalue count (4 forward-looking variables) and steady state unchanged. Figure 2 sweeps $(b_\pi, b_y) \in [0, 5] \times [0, 3]$ on a 60×60 grid – the b_π range extended from the earlier rounds’ $[0, 3]$ specifically so the current round’s posterior mean, which the diffuse prior moved to $b_\pi = 3.474$, actually falls inside the plotted region rather than off its edge. Of the 3,600 grid points, 2,857 are simultaneously determinate and E-stable. The determinate and E-stable classifications were verified to coincide exactly, cell-for-cell, with zero mismatches across all 3,600 grid points, as expected from Proposition 10.3 for this class of model. Because the two conditions are literally identical here, they are combined into a single two-region figure rather than shown as separate determinacy and E-stability panels. At the posterior mean $(b_\pi, b_y) = (3.474, 0.083)$ the model is determinate and E-stable – comfortably inside the determinate region, since the grid’s own determinacy threshold along the $b_y = 0.083$ line is $b_\pi \approx 1.10$ and the posterior mean sits more than three times that value.

As a robustness check on the fiscal feedback parameter φ_b , determinacy and E-stability were re-evaluated at $\varphi_b \in \{0.0301, 0.0689, 0.1005\}$ (the current run’s 90% HPD lower bound, posterior mean, and HPD upper bound), holding (b_π, b_y) at their posterior means:

φ_b	Determinate?	Spectral radius	E-stable?
0.0301 (HPD lower)	Yes	0.9948	Yes
0.0689 (posterior mean)	Yes	0.9837	Yes
0.1005 (HPD upper)	Yes	0.9744	Yes

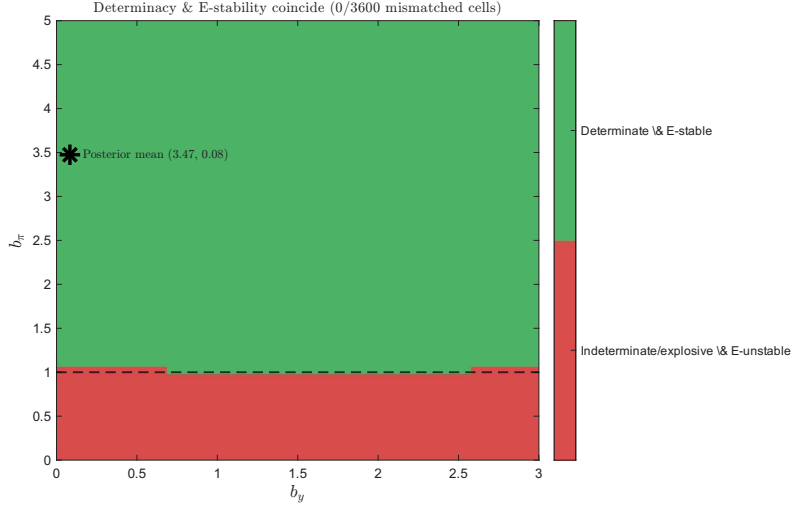


Figure 2: Combined determinacy/E-stability region over $(b_\pi, b_y) \in [0, 5] \times [0, 3]$, all other parameters at posterior means (current, diffuse-prior round). Green: determinate *and* E-stable (the two conditions coincide exactly on this grid, with 0/3,600 mismatched cells). Red: indeterminate or explosive, and E-unstable. The star marks the posterior mean $(b_\pi, b_y) = (3.474, 0.083)$, now visible within the plotted range.

Stability holds at all three points, with the spectral radius falling (i.e., stability strengthening) as φ_b rises across this range, as in every previous round. The spectral radii are somewhat closer to the instability boundary (1.0) than in the previous round (which ranged 0.9674–0.9844 across its φ_b HPD), consistent with this round’s lower φ_b HPD range sitting closer to zero, but stability is not in question at any of the three points.

7 Steady-State Annualized Rates

Table 9 reports the model’s steady-state gross quarterly and annualized rates evaluated at the full posterior mean parameter vector (Table 6, current diffuse-prior round), obtained by re-solving the non-ZLB (RELAX) steady state with `resol()` after loading the posterior means – necessary because v^a, η (which do enter the steady state, unlike $b_\pi, b_y, a_i, b_r, \rho_z, \varphi_b$, which affect only the model’s log-linear dynamics around it) moved from the previous round. As a consistency check, the steady-state real rate satisfies the Euler equation exactly: $r^\ell/\pi = 1/\beta = 1.003311$.

These are close to, but not identical to, the previous round’s steady state (3.79% vs. 3.63% inflation, 1.86% vs. 1.74% deposit rate) – a modest shift consistent with v^a and η ’s posterior means moving somewhat (Table 6 vs. Table 7) while $\bar{\tau}, \chi, \beta$ and every other steady-state-relevant calibrated parameter are unchanged; the reserve-equation fix itself leaves the steady state exactly unchanged (verified directly – see Section 1). Both qualitative features noted in the previous round persist. First, steady-state annual inflation (3.63%) remains well above the original ($\bar{\tau} = 0.20, \chi = 1.0$) calibration’s 1.10%, consistent with the larger calibrated deficit ($\bar{\tau}$ lowered from 0.20 to 0.14) dominating the lower χ ’s inflation-reducing effect. Second, the steady-state deposit rate r^d remains positive (1.74% annualized), as in the previous round, in contrast to the fractionally negative values seen under the original calibration. The discount factor $\beta = 0.9967$ was originally chosen so that the loan rate would clear a 2% inflation target while keeping

the deposit rate above the ZLB; under the recalibrated $\bar{\tau}, \chi$ that margin remains comfortably positive.

Symbol	Description	Value
<i>Household / preferences</i>		
β	Discount factor	0.9967
χ	Shopping-time curvature	0.5 (recalibrated from 1.0)
ψ	Inverse Frisch elasticity	1.3
z	Long-run labor productivity	1.3
<i>Banking sector</i>		
ν	Deposit CES elasticity	0.25
x_n	Loan portfolio share	0.95
$x_h = 1 - x_n$	Deposit-labor portfolio share	0.05
ϕ_n	Reserve management cost	0.0005
x_a	Deposit-technology level (steady state)	5.2528
<i>Price setting</i>		
θ	Goods demand elasticity	6.0
ϕ_p	Rotemberg price-adjustment cost	50
<i>Fiscal / debt</i>		
g	Steady-state govt. spending (share of output)	0.20
$\bar{\tau}$	Steady-state tax target	0.14 (recalibrated from 0.20)
$g - \bar{\tau}$	Implied steady-state deficit/GDP	6% (from $\approx 0\%$)
b_0	Steady-state debt target	1.0
ω	Fraction of bonds held by the central bank	0.0
<i>Nominal anchors</i>		
π^{ss}	Reference gross quarterly inflation (2% annual)	1.004963
$r^{\ell,ss}$	Reference gross quarterly loan rate (Fisher: π^{ss}/β)	1.008290
$\underline{r}^n$	ZLB floor (gross quarterly rate)	1.0
<i>Shock persistence (fixed, not estimated)</i>		
ρ_a	TFP persistence	0.95
ρ_{xa}	Deposit-technology persistence	0.95
ρ_{va}	Preference-shock persistence	0.0
ρ_g	Govt. spending persistence	0.0
ρ_τ	Tax-rule persistence	0.5

Table 5: Calibrated (non-estimated) parameters for the results in this note. π^{ss} and $r^{\ell,ss}$ are reference values used only in the tax-rule/debt-issuance block (Section 1) and the Taylor-rule normalization; the model’s actual *endogenous* non-ZLB steady state, reported in Table 9, need not equal them exactly. ρ_a, ρ_{xa} are fixed rather than estimated because freeing them previously drove ρ_a to a near-unit-root value and produced a non-positive-definite Hessian at the mode (Section 2.2).

Parameter	Prior mean	Post. mean	90% HPD	Prior
v^a (preference scale)	2.500	2.890	[2.055, 3.482]	Gamma(2.50, 0.75)
η (labor preference)	1.530	1.136	[0.922, 1.326]	Gamma(1.53, 0.40)
b_π (inflation response)	1.000	3.474	[3.344, 3.604]	Gamma(1.00, 1.00)
b_y (output response)	1.000	0.083	[0.057, 0.106]	Gamma(1.00, 0.70)
a_i (policy-rate level weight)	0.490	0.181	[0.169, 0.195]	Beta(0.49, 0.10)
b_r (MP shock persistence)	0.360	0.262	[0.171, 0.347]	Beta(0.36, 0.10)
ρ_z (productivity persist.)	0.450	0.443	[0.292, 0.549]	Beta(0.45, 0.10)
φ_b (fiscal feedback)	0.050	0.069	[0.030, 0.101]	Beta(0.05, 0.05)
<i>Standard deviation of shocks</i>				
σ_a	0.050	0.017	[0.016, 0.018]	InvGamma(0.05, 0.025)
σ_z	0.050	0.023	[0.018, 0.028]	InvGamma(0.05, 0.025)
σ_{va}	0.050	0.039	[0.026, 0.050]	InvGamma(0.05, 0.025)
σ_{xa}	0.050	0.036	[0.032, 0.041]	InvGamma(0.05, 0.025)
σ_g	0.050	0.030	[0.025, 0.036]	InvGamma(0.05, 0.025)
σ_r	0.050	0.024	[0.021, 0.026]	InvGamma(0.05, 0.025)
σ_τ	0.050	0.031	[0.024, 0.036]	InvGamma(0.05, 0.025)
σ_L (COVID labor-supply)	0.150	0.069	[0.062, 0.076]	InvGamma(0.15, 0.075)
$\sigma_{me,y}$	0.050	0.021	[0.018, 0.023]	InvGamma(0.05, 0.025)
$\sigma_{me,d}$	0.050	0.018	[0.015, 0.022]	InvGamma(0.05, 0.025)
$\sigma_{me,\pi}$	0.050	0.014	[0.013, 0.014]	InvGamma(0.05, 0.025)
$\sigma_{me,r}$	0.050	0.010	[0.009, 0.012]	InvGamma(0.05, 0.025)
σ_{me,r^n}	0.010	0.003	[0.002, 0.003]	InvGamma(0.01, 0.005)
$\sigma_{me,\tau}$	0.050	0.020	[0.017, 0.024]	InvGamma(0.05, 0.025)
$\sigma_{me,bT}$	0.050	0.022	[0.019, 0.025]	InvGamma(0.05, 0.025)

Table 6: Posterior estimates, current (diffuse-prior) round: 2 chains $\times$ 50,000 draws, `mh_jscale=0.05`, `hessian` jumping covariance, true OccBin ZLB likelihood, reserve-equation fix (Section 1), FYGFDPUN-based debt observable, $\bar{\tau} = 0.14$, $\chi = 0.5$. ρ_a, ρ_{xa} remain fixed at 0.95 (see Section 2.2).

Parameter	Prior mean	Post. mean	90% HPD	Prior
v^a	2.500	2.776	[1.526, 3.830]	Gamma(2.50, 0.75)
η	1.530	0.988	[0.666, 1.266]	Gamma(1.53, 0.40)
b_π	2.100	2.169	[2.015, 2.315]	Normal(2.10, 0.15)
b_y	0.280	0.135	[0.068, 0.188]	Gamma(0.28, 0.10)
a_i	0.490	0.180	[0.161, 0.199]	Beta(0.49, 0.10)
b_r	0.360	0.232	[0.151, 0.333]	Beta(0.36, 0.10)
ρ_z	0.450	0.562	[0.410, 0.720]	Beta(0.45, 0.10)
φ_b	0.100	0.094	[0.066, 0.124]	Beta(0.10, 0.05)

Table 7: Previous round’s posterior estimates (structural parameters only), reproduced for comparison: pre-fix reserve equation, tight b_π, b_y, φ_b priors, custom pooled-posterior jumping covariance, `mh_jscale=0.25`.

Parameter	Chain 1 p	Converged?	Chain 2 p	Converged?
v^a	0.350	Yes	0.558	Yes
η	0.088	Yes	0.008	No
b_π	0.062	Yes	0.089	Yes
b_y	0.281	Yes	0.070	Yes
a_i	0.024	No	0.563	Yes
b_r	0.001	No	0.001	No
ρ_z	0.564	Yes	0.000	No
φ_b	0.226	Yes	0.081	Yes

Table 8: Geweke convergence test p -values, 15% taper, by chain (current diffuse-prior round, `hessian` covariance, `mh_jscale`=0.05, $\bar{\tau} = 0.14$, $\chi = 0.5$).

Symbol	Description	Gross quarterly	Annualized (%)
π	Inflation	1.008964	3.63
r^ℓ	Loan rate	1.012304	5.01
r^d	Deposit rate	1.004333	1.74
r^n	Policy rate (interest on reserves)	1.002214	0.89

Table 9: Steady-state gross quarterly rates and simple annualized percentages $((\text{rate}^4 - 1) \times 100)$ at the posterior mean parameter vector, non-ZLB (RELAX) regime, $\bar{\tau} = 0.14$, $\chi = 0.5$, current (diffuse-prior) round.